Инфраструктура и измерения

Compose → readiness → воспроизводимость → время/RAM/диск/рост

Что нужно знать

- Порядок db healthy → loader success → pipeline success → app. Почему depends_on с успешным завершением отличается от простого запуска процесса.
- Named volume базы и bind mounts для CSV, модели и измерений; почему секреты и сырые данные исключены из Git и Docker build context.
- Методика: три повторения, среднее и отдельные результаты, одна изменяемая величина, характеристики ноутбука.
- Полный запуск 31.10 s относится к готовым CSV и образам; это не время первого скачивания и сборки.
- Индекс сравнивается на одной выборке 300 клубов; 25/50/100% эксперименты — отдельная агрегация крупнейшей таблицы.
- Память — максимальное наблюдаемое working set из docker stats, а не непрерывно измеренный истинный пик.
- Что нужно подготовить к защите: данные и образы заранее, рабочий ноутбук, сохранённые результаты, PDF и двухминутное резервное видео.

Что показать

1. Открыть docker-compose.yml и проследить зависимости и mounts без публикации паролей.
2. Показать README/MEASUREMENTS.md: машина, три прогона, диапазон условий и рост данных.
3. Объяснить отличие 31.10 s запуска от 10.872/0.0748 s запросов и от времени импорта одного CSV.

Короткая речь на английском

> *Docker Compose coordinates four core services. The database must be healthy, loading and processing must finish successfully, and then the application starts. Source files, the model and measurement results use explicit mounts. We measured timing, observed memory, disk footprint and growth in separate experiments, each repeated three times. Complete cached-source startup averaged 31.10 seconds on an Apple M3 Pro with 18 GiB RAM. The selective index lookup improved by about 145 times, but that result applies to that query and sample. Caches were retained, image builds were excluded and brief memory peaks may have been missed.*

Переход: “Taken together, the system is reproducible on the measured setup, while the data and model limitations remain explicit.”

Твой тайминг на защите

Слайд	Блок	Время	Длительность
8. Measure the whole system	08 / Performance	10:15–12:30	135 с
9. Growth, memory and failure cases	09 / Reliability	12:30–14:00	90 с

Как закрывать экзамен

1. Твои слайды 8 и 9 (10:15–14:00) — ты закрываешь содержательную часть, последнюю минуту берёт Джакоппо, и твоя часть прямо соответствует обязательному пункту «производительность в числах».
2. Каждое число произноси с условиями: 31.10 с — среднее трёх прогонов с готовыми источниками и собранными образами, без скачивания и без сборки.
3. 145.4× называй как результат одного эксперимента на 300 выборочных запросах, а не как ускорение всего приложения. Этот вопрос задают почти всегда.
4. Память описывай методом: максимальная наблюдённая выборка docker stats с интервалом около двух секунд, не гарантированный пик.
5. Рост 25 / 50 / 100% объясняй нелинейностью: 4× строк дали 4.40× загрузки и 3.35× агрегации из-за фиксированных затрат, планов и кэшей.
6. Меняй один фактор за раз — скажи это вслух, это прямое требование протокола измерений из брифа.
7. В конце назови, что осталось: лицензия, независимый запуск на другом ноутбуке, резервное видео и фактический состав участников. Не заявляй их выполненными.

Где смотреть в проекте

Файл в проекте	Что в нём смотреть
`docker-compose.yml`	Порядок запуска: healthcheck базы → успешный loader → успешный pipeline → приложение. Профили для measure и legacy UI.
`docs/MEASUREMENTS.md`	Три прогона, железо (M3 Pro / 18 GiB), метод выборки памяти и все оговорки. Главный документ твоей части.
`docs/ARCHITECTURE.md`	Контейнеры как блоки, стрелки как файлы, строки, SQL и HTTP — структура для слайда архитектуры.
`docs/COURSE_REQUIREMENTS.md`	Таблица требований курса со страницами брифа и статусом каждого пункта.

Файл в проекте	Что в нём смотреть
`measure/benchmark.py`	Эксперимент с индексом: 300 одинаковых выборочных запросов, 145.4× — и почему это не ускорение всего приложения.
`scripts/measure_system.py`	Полный прогон с нуля: старт 31.10 с, loader 12.53 с, pipeline 9.7 с (средние трёх прогонов).
`scripts/report_measurements.py`	Сборка измерений в отчёт — показывает, что числа не вписаны руками.
`docs/AUDIT.md`	Найденные и исправленные дефекты: ответ на вопрос «как вы убедились, что это работает».
`docs/WORK_PACKAGES.md`	Пакеты работ, приёмочные условия и честный список незакрытых внешних пунктов.
`scripts/build_defense.py`	Сборка этого пособия: слайды, роли, PDF и проверка, что выступление ровно 900 секунд.

Словарь простыми словами

Определения намеренно нетехнические: ими удобно объяснять систему человеку вне курса и отвечать на вопрос, формулировку которого ты не узнал(а).

Container / Docker Compose	A container is one packaged program with everything it needs to run. Compose is the file that starts our four containers in the right order: the database first, then the loader, then the processing pipeline, then the web application.
Relational database / PostgreSQL	A store of tables with fixed columns that can be combined by shared keys. We use PostgreSQL 16. Combining two tables on a shared key is called a join.
Schema	The declared shape of the tables: which columns exist and what type each one holds. It is written in sql/schema.sql.
COPY and index	COPY loads a whole file into a table in one stream. An index is a lookup structure, like the index of a book: it makes selective queries fast and is built after the bulk load.
Pipeline	The processing step between the database and the web page. It reads tables, computes our measures and the model, and writes prepared result tables back.
API / endpoint / JSON	The API is the set of web addresses the page calls. Each address is an endpoint. JSON is the plain-text format of the answer.
Champagne Index	Our own measure of attacking output: recorded goals plus assists, divided by the number of matches the source reports for that club-season. It is a proxy, not an official statistic.
Target and feature	The target is what the model tries to predict — here the change in the index in the next consecutive season. A feature is an input the model uses to predict it.

Chronological holdout	We train the model on earlier seasons and test it on later seasons it has never seen. A random split would mix the future into the training data and flatter the result.
Baseline	The simplest possible prediction, used for comparison. Ours predicts no change at all. A model is only useful if it beats the baseline on the same test rows.
R² and MAE	R² is the share of variation the model explains, from 0 to 1 — ours is 0.127, which is weak. MAE is the average absolute error in index units — ours is 0.4856 against a baseline of 0.5114, about 5% better. Neither is a percentage of correct answers.
Correlation vs. causation	A correlation (r) measures how two numbers move together, from −1 to +1. It cannot show that one causes the other. Every result in this project is a correlation, including the red-card and spending findings.
Proxy	A measurable stand-in for something we cannot measure directly. Our shots-on-target estimate is a proxy for chance quality; it is not xG, which the data source does not contain.
Poisson rating	A simple statistical model of how many goals each team scores and concedes, fitted to one league-season. It describes that season; it does not forecast new matches.
SHAP	A method that attributes a model's prediction to its individual inputs, so we can report which feature mattered most without relying on the model's internal importance score.

Практические задания

1. По схеме объясни, что произойдёт, если loader завершится с ошибкой, а база уже healthy.
2. Сравни 25% и 100%: строк в 4 раза больше, загрузка примерно в 4.40 раза дольше; объясни, почему это не универсальный закон.
3. Подними проект по README на другом ноутбуке и запиши наблюдения. Не объявляй эту проверку выполненной до реального запуска.

Вопросы по направлению

What starts when you run Docker Compose?

Что запускает Compose?

The database becomes healthy, then the loader must finish successfully, then the processing pipeline must finish, and finally the application starts. The measurement service is on demand and the legacy interface uses an optional profile.

What does the 31.10-second measurement include?

Что входит в 31.10 секунды?

A mean over three complete starts with prepared source files and existing images: containers are recreated, sources reimported and the model retrained until the API is ready. Network acquisition and image builds are excluded and caches are not flushed.

Does the index make the whole application 145 times faster?

Всё приложение быстрее в 145 раз?

No. The measured gain is for 300 identical selective club lookups on the raw performance table, comparing index conditions. The frontend reads prepared outputs and a full GROUP BY has a different access pattern.

Is your memory number a guaranteed peak?

Это гарантированный пик RAM?

It is the largest observed Docker working-set sample during three runs, sampled at roughly two-second intervals. Brief peaks may be missed. We report the method rather than label it a continuous or hardware-independent maximum.

Why did four times as many rows not give exactly four times the query time?

Почему масштабирование не точно линейное?

Fixed costs, query plans, indexing, caches and hardware influence scaling. In this experiment four times the largest-table rows gave about 4.40 times the loading time and 3.35 times the aggregation time. That does not define a general law.

What remains before a group submission?

Что ещё нужно для сдачи?

Confirm actual team membership and each person’s contribution, run an independent second-laptop setup, resolve the source licence, rehearse the presentation and record a backup demo. Equal author email counts alone do not demonstrate real group participation.

What is the main result in one sentence?

Главный результат одним предложением?

The system is reproducible on the measured setup, while historical transfer spending alone has almost no linear relationship with next-season index change and the richer model adds only modest predictive value.

What is the most important limitation?

Главное ограничение?

The data are retrospective and unevenly covered, the index is a proxy, and associations are not causal effects. Source licence and independent team/laptop validation also remain explicit unresolved requirements.

How do goals for, goals against and goal difference work in the comparison?

Как в сравнении рассчитываются забитые, пропущенные голы и разница?

GF and GA are season totals rebuilt from completed home/away match scores. GD is GF − GA. They come from the optional football-data.co.uk match enrichment, not from the Champagne Index player totals.

How are attacking and defensive ranks calculated?

Как рассчитываются атакующий и защитный рейтинги?

Within one league and season, attack rank orders goals scored per match from highest to lowest; defense rank orders goals conceded per match from lowest to highest. SQL RANK gives tied rates the same rank and leaves a gap after ties. They describe scores, not tactics or chance quality.

Do we have xG or can we estimate it?

Есть ли в данных xG или можем ли мы его оценить?

The source has no provider xG field. The comparison instead shows observed shots and shots on target. A separate estimate multiplies shots on target by the league-season average goals-per-shot-on-target rate; it is an experimental volume proxy with no shot-location or chance-quality input, not xG.

What is the difference between the Champagne Index and attack rank?

Чем атакующий рейтинг отличается от Champagne Index?

The Champagne Index is recorded player goals plus assists divided by a Transfermarkt matches proxy for a club-season. Attack rank uses actual completed league scorelines, sums team goals and ranks goals per match within the league-season. Their sources, coverage and units differ.

How did you check the goalkeeper defensive totals?

Как вы проверили вратарские показатели по пропущенным?

We compared Transfermarkt goalkeeper goals-conceded totals with the independent match-result totals on 388 adequately covered top-five club-seasons. They match exactly in 366 (94.3%). Clean sheets match only 269 (69.3%), so the scoreline source remains the basis of league ranks.

What does the defensive-spend analysis show?

Что показывает анализ расходов на защитников?

The Pearson correlation between defender transfer fees and next-season change in the goalkeeper-source conceded-goal rate is r=0.004047, p=0.426997, n=38,531. The sample does not support a useful linear association. It is descriptive, not causal. A separate match-level check finds teams with a red card concede 0.341 more goals on average in the same match, but match context confounds that association.

Why keep the player-source defensive values separate from scoreboard goals?

Почему defensive-показатели по игрокам отделены от итогов матчей?

The sources use different records and denominators. Goalkeeper conceded totals agree with scorelines in 94.3% of adequately covered seasons, while clean sheets agree in 69.3%. The UI therefore uses match scores for GF, GA, GD and ranks; player-source totals remain a quality cross-check. Shots and corners are also observed match statistics, not chance-quality measures.

Is the shot-based estimate the same as xG?

Оценка по ударам — это xG?

No. It multiplies a team’s shots on target by the league-season average goals per shot on target. It has no shot-location, distance, body-part, pressure or goalkeeper-position input. It is a rough volume benchmark and must be called an SOT-based proxy, not provider xG.

What does the Poisson attack/defense rank tell us?

Что показывают Poisson-ранги атаки и обороны?

The model fits attack, defense and a shared home-advantage term to the selected league-season scores. Larger fitted attack and defense parameters mean stronger estimated attack and defense. Its ranks are descriptive and in-sample; the separate chronological RandomForest evaluation is the project’s held-out prediction check.

Did the added defensive features improve the model?

Улучшили ли защитные признаки модель?

On the same chronological split, R^2 moves from 0.1271 with the five-feature baseline to 0.1502 after adding goalkeeper conceded rate, discipline and net spend; MAE moves from 0.4856 to 0.4788 on 5,585 held-out rows. The gain is modest, and it does not isolate a causal effect of defense.

Do red cards cause more goals to be conceded?

Красные карточки приводят к большему числу пропущенных?

The local sample shows a same-match difference of +0.341 conceded goals per team-match (1,619 sent-off vs 16,195 others; Welch $p < 0.0001$). This is observational: stronger opponents, match state and timing can affect both red cards and goals. It is not a causal estimate.

Какие фразы не использовать

- 31 секунда включает скачивание 353 MiB и сборку образов.
- docker stats гарантирует истинный максимальный расход памяти.
- Одинаковое число коммитов подтверждает одинаковый личный вклад.

Экзаменационный день

Дата	Формат	Что требуется
12 November 2026 CIDS Lecture 15, pp. 37–39	Checkpoint · 5 minutes	One end-to-end demo, one visible result and the first measured full-run time. A polished interface is not required.
20 / 26 November 2026 CIDS Lecture 15, pp. 37–39	Final defense · 15 minutes + 5 minutes of questions	Every real participant speaks. Suggested split: question and data 3 min, architecture 2, live demo 3, measurements 3, problems 2, contributions 2.

Подготовка машины и материалов

1. Thirty minutes ahead: run `docker compose up -d --build`, then open /api/health and wait for `ready=true` with the loader and pipeline finished.
2. Open four tabs in advance: Overview (Serie A 24/25), the Inter / Milan comparison, the Lab, and the Project page with the numbers.
3. Open the offline study site `defense/site/index.html` and `defense/site/downloads/Presentation.pdf`: both work without a network and without the database.
4. Check that the screenshots in `defense/site/assets/*-desktop.png` are present. They are the fallback if Docker does not come up.
5. Turn off notifications, connect the charger, test screen mirroring and increase the browser font size.
6. Decide in advance who drives the demo and who watches the clock. The timer is built into `defense/site/presentation.html` (N for notes, F for fullscreen).

Как отвечать на вопросы

- Answer in three steps: one short sentence, then the evidence (a file or a number), then the limitation.
- Give every number with its sample and unit: “r = −0.79 across 388 club-seasons, within the same season”, not “the correlation is high”.
- If you do not know, say where the answer lives instead of guessing: “that is in docs/MEASUREMENTS.md, the memory sampling method is described there”.
- If the question belongs to someone else's part, hand it over in one sentence rather than approximating their area.
- Never turn an association into a cause. Say “accompanies” or “is associated with”, never “causes” or “improves”.
- If asked about a weak point, answer it first and directly. An acknowledged limitation is stronger than a defended inaccuracy.

Неудобные вопросы: отвечай прямо и первым

Тема	Вопрос	Ответ и где подтверждение
Licence of the primary source	Is your data openly licensed?	We pinned the source revision and verified headers and checksums, but there is no licence file, so we did not establish an explicit reuse licence. We document it as unresolved and we do not redistribute the raw data. Public downloadability is not a licence. `docs/SOURCES.md, docs/LICENCE_DECISION.md`
Four contributors	Who did what in this project?	Describe only the work you can open and explain on screen. Git identities and role cards are not proof of independent contribution, and the repository says so. `docs/WORK_PACKAGES.md, README.md`
Reproduction on another machine	Did someone else reproduce it?	The run is reproducible on the measured machine, and we also rebuilt it from a clean clone. An independent fresh checkout on another person's laptop has not been recorded yet, so we do not claim it. `docs/FRESH_CHECKOUT.md, docs/MEASUREMENTS.md`
xG and proxies	Is this xG?	No. The source has no provider xG field. We show observed shots and shots on target, and a shots-on-target conversion proxy that we label experimental. The Poisson attack and defense ratings describe the season they are fitted on. `docs/GLOSSARY.md, pipeline/proxy_xg.py, pipeline/power_ratings.py`

> Роль — учебная ответственность. Не заявляй личное авторство или самостоятельную проверку, если этого не было. Покажи ровно те изменения и команды, которые выполнял(а) лично.